

Dilations & Reflections (The “a” Value)

The Big Idea

The a value controls TWO things: **how steep** the graph is and **which direction** it goes.

Cubic Form	Cube Root Form	Rational Exponent Form
$f(x) = a(x - h)^3 + k$	$f(x) = a\sqrt[3]{x - h} + k$	$f(x) = a(x - h)^{\frac{1}{3}} + k$

What Does “a” Do?

If...	Then the graph...	Example
$ a > 1$	is stretched (steeper, taller)	$a = 2$: twice as steep
$ a < 1$	is compressed (flatter, shorter)	$a = \frac{1}{2}$: half as steep
$a < 0$	is reflected (flipped over x -axis)	$a = -1$: upside down

How to apply: Multiply every y -coordinate by a .

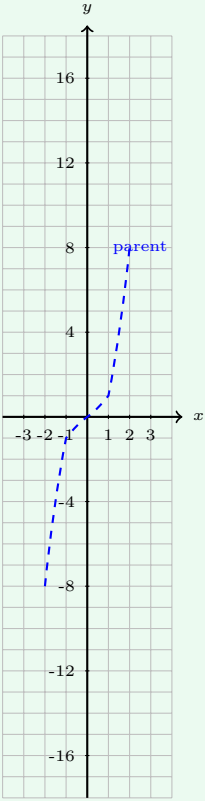
Example 1: $f(x) = 2x^3$ (Vertical Stretch)

$a =$ _____

Effect: _____

Multiply each y by 2:

Parent	$y \times 2$	New Point
$(-2, -8)$	$-8 \times 2 =$	$(-2, \quad)$
$(-1, -1)$	$-1 \times 2 =$	$(-1, \quad)$
$(0, 0)$	$0 \times 2 =$	$(0, \quad)$
$(1, 1)$	$1 \times 2 =$	$(1, \quad)$
$(2, 8)$	$8 \times 2 =$	$(2, \quad)$



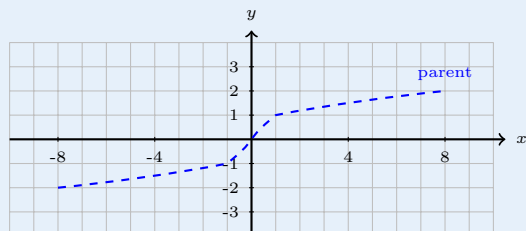
Example 2: $f(x) = -\sqrt[3]{x}$ (Reflection)

$a =$ _____

Effect: _____

Multiply each y by -1 :

Parent	$y \times (-1)$	New Point
$(-8, -2)$	$-2 \times (-1) =$	$(-8, \quad)$
$(-1, -1)$	$-1 \times (-1) =$	$(-1, \quad)$
$(0, 0)$	$0 \times (-1) =$	$(0, \quad)$
$(1, 1)$	$1 \times (-1) =$	$(1, \quad)$
$(8, 2)$	$2 \times (-1) =$	$(8, \quad)$



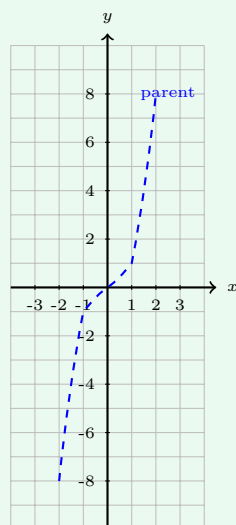
Example 3: $f(x) = -\frac{1}{2}x^3$ (Compression + Reflection)

$a =$ _____

Effect: _____

Multiply each y by $-\frac{1}{2}$:

Parent	$y \times (-\frac{1}{2})$	New Point
$(-2, -8)$	$-8 \times (-\frac{1}{2}) =$	$(-2, \quad)$
$(-1, -1)$	$-1 \times (-\frac{1}{2}) =$	$(-1, \quad)$
$(0, 0)$	$0 \times (-\frac{1}{2}) =$	$(0, \quad)$
$(1, 1)$	$1 \times (-\frac{1}{2}) =$	$(1, \quad)$
$(2, 8)$	$8 \times (-\frac{1}{2}) =$	$(2, \quad)$



Quick Summary

$a = 2$	$a = \frac{1}{2}$	$a = -1$	$a = -2$
Steeper	Flatter	Flipped	Flipped AND steeper

Remember: The sign of a controls direction. The size of $|a|$ controls steepness.